Apply non-linear filters (median or other rank filters) to remove noise from data/images provided in OPE7 and perform comprehensive analysis. **CO3, CO4**

Identify noise types present in the dataset (salt-and-pepper, impulse, speckle, etc.).

Implement median filtering with varying window sizes and, optionally, other non-linear filters (alpha-trimmed mean, morphological filters).

Analyze how filter size and type affect detail preservation vs. noise removal.
Observations: optimum filter sizes per noise type, edge preservation analysis, and failure cases.
Summarize best-performing methods for each noise type and practical recommendations.







